Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

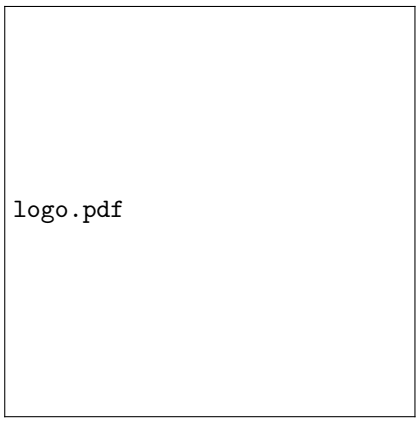
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Department of Computer Science
Doctoral administration

ETH Zurich
CAB H 37.1
Universitätstrasse 6
CH-8092 Zurich

doctorate@inf.ethz.ch
www.inf.ethz.ch

Guidelines for Doctoral Plan

This document aims to help you write your doctoral plan, which you have to submit in advance of the aptitude colloquium. An electronic copy must be submitted within eleven months after provisional admission to the D-INFK doctoral coordination.

The integrated research proposal will be the basis of your aptitude colloquium, which you need to pass in order to advance to definitive admission as a doctoral student.

Please note that the doctoral plan has not the character of a legally binding agreement but is instead a declaration of intent between you and your supervisor and should be conceived as a living instrument. Changes are possible, and in many cases, certainly necessary. Major changes should be documented in the annual status conversations.

Basic Information

Name Doctoral Student _____

Student Number _____

Name Supervisor _____

Name Second Advisor _____

Start of Doctorate _____

Title Research Proposal

Part 1 Research Proposal

(please add extra pages here; use the following structure to describe your research work)

1. Abstract

Present the background of your research project, list its main goals, describe the methods to be used and the expected results, as well as their impact on the field and beyond.

2. Introduction and Current State of Research in the Field

Describe your project in the context of the current state of knowledge in your field. Refer to the most important publications. Describe briefly:

- *Which previous insights provided the starting point and basis for your project*
- *In which areas research is needed, and why*

3. Goals of the Thesis

Describe the research questions you address in your research project and the concrete objectives that you aim to achieve in the period of your doctorate.

Illustrate the expected impacts of your project: how will your proposed research add to existing knowledge in the field? How and where might your results be used? How do you intend to disseminate your research?

4. Progress to Date

Present the research work you have already undertaken in the relevant field, describe the results obtained so far (including publications), as well as the relevance of these preliminary undertakings for your project. Also, describe your "negative results": what could you learn from this if something went wrong?

5. Detailed Work Plan

Describe the studies or experiments envisaged to reach the set goals. Present your work plan in the form of work packages (WP), each defining the task and describing the methods used and the expected outcome. In general, a work package should describe 3 - 12 months of work. Assess the risks involved for each work package: if the work package does not work, do you have various options, which other parts of your plan are affected, what is the backup?

WP 1: ...

WP 2: ...

6. Publication Plan

For each work package, list the tentative publication in a specific conference or journal.

7. Time Schedule (potentially as a visualization)

Indicate the tentative timeframe for each work package. Your time schedule should be based on the average time of 4 years of doctorate.

8. References

List the sources of all work referred to in the research proposal. Give the full reference, especially the title, source, and full author list (use a style commonly adopted in your research field).

Part 2 Teaching Responsibilities

Describe which teaching tasks you are assigned to and indicate (roughly) how many hours you devote to such activities each week.

Teaching tasks are, e.g., assisting in lectures, practical classes, exercise classes, correcting students' reports, supervising undergraduates in semester projects, summers schools, marking/overseeing exams.

Part 3 Other Duties

If relevant, describe which additional tasks you are assigned to (e.g., operation or maintenance of equipment). Indicate (roughly) how many hours you devote to such activities each week.

Part 4 Study Plan

Please add your individual study plan here, including a tentative schedule (e.g., using the learning agreement in mystudies). ***NOTE****: a course in ethics and good scientific practice is a* ***mandatory*** *requirement.*

Signatures

Supervisor _____

Second Advisor _____

Doctoral Student _____

Date _____